Section A :

Question I :

1. We decided to use the variable **"Politicians don't care what people like me think"** as the variable expressing people's level of satisfaction with political parties or politicians, since it reflects how people feel about the way politicians treat them. It can therefore be translated as the level of concern the politician feels about his or her concerns.

   A variable that reflects the respondent's level of income is "**Income - gross household (represents the total income of your household from all sources before tax - including benefits, savings and so on?)".**

   A variable representing the voter's attitude to voting is **"Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election?"** because it expresses whether the respondent is in the habit of voting. Its association with the other two could explain whether the respondent's refusal to vote is due to his level of satisfaction with politicians or his income level.

2. **Review :**

- **Variable 1: Politicians don't care what people like me think**

This variable has 05 measurement levels, ranging from "1 = Strongly disagree" to "5 = Agree" to "3 = Neither disagree nor agree", plus the options "-999 = Not stated" and "-1 = Don't know".

These measurement levels provide a little more information. But I think the number could have been reduced to make it easier to interpret. For example, the modalities "Disagree and Totaly disagree" or "Agree and Totaly agree" could have been only two modalities instead of four.

- **Variable 2: Income - gross household (represents the total income of your household from all sources before tax - including benefits, savings and so on?)**

This variable has 15 measurement levels, ranging from "1 = Under GBP 2,600" to "15 = GBP 100,000 or more", plus the options "-999 = Not stated", "-2 = Refused" and "-1 = Don't know".

The large number of revenue categories provides more specific information, but does not facilitate an overall interpretation of the results. As the saying goes, too much information takes us away from the essential.

Also, the "Refused" and "I don't know" modalities could have been represented as one. For a person who declares that he or she doesn't know his or her income level could be seen as someone who doesn't want to divulge this information.

- **Variable 3: Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election?**

This variable has just four possibilities: "-999 = Not stated", "-2 = Refused", "-1 = Don't know", "1 = Yes, voted" and "2 = No, did not vote".

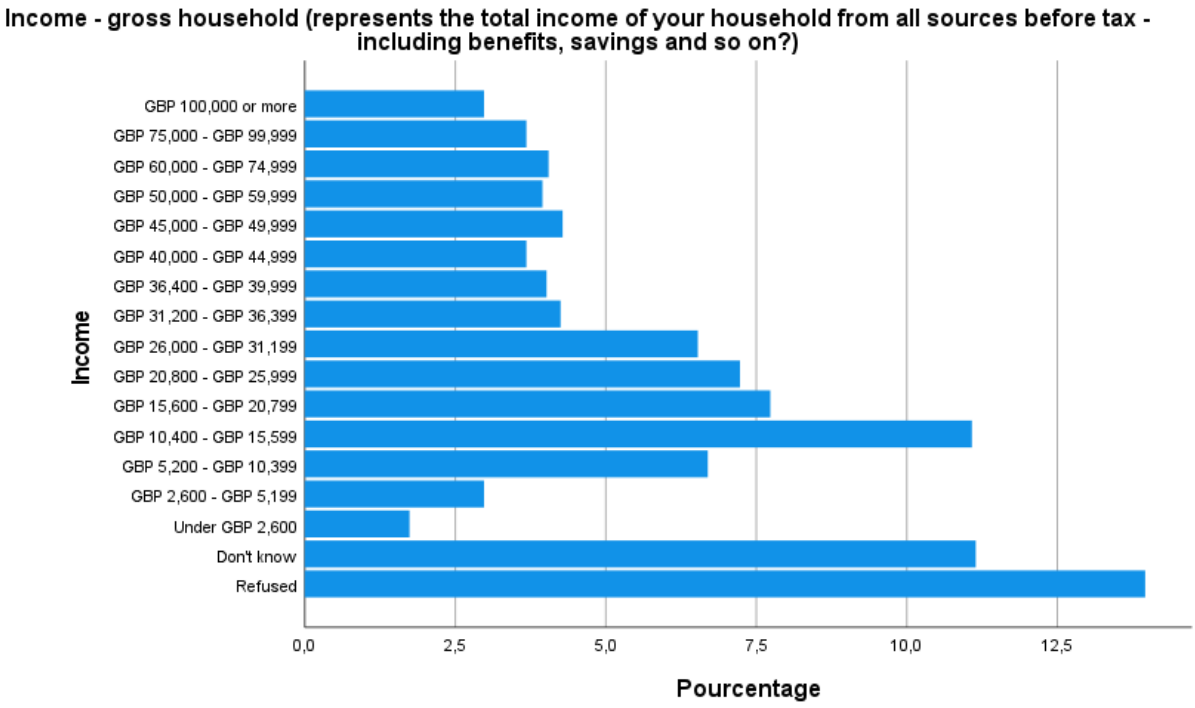
Here, two important modalities: "Yes, voted" and "No, did not vote". In addition, as in the case of variable 2, a person who doesn't know whether they voted or not can be equated with someone who doesn't want to talk about it. So reduce the "I don't know" modality.

3. The variables we have chosen have already been encoded in the database to facilitate the calculation of the most common descriptive statistics. However, this encoding will not be used in the tables and graphs, to allow the reader to directly understand the information presented, without having to refer to a dictionary of encodings before being able to understand the information represented.

We have tried to give an overview of the encoding used for the three variables in the previous question.

Income - gross household (represents the total income of your household from all sources before tax - including benefits, savings and so on?)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Refused	417	14,0	14,0	14,0
	Don't know	333	11,1	11,1	25,1
	Under GBP 2,600	52	1,7	1,7	26,8
	GBP 2,600 - GBP 5,199	89	3,0	3,0	29,8
	GBP 5,200 - GBP 10,399	200	6,7	6,7	36,5
	GBP 10,400 - GBP 15,599	331	11,1	11,1	47,6
	GBP 15,600 - GBP 20,799	231	7,7	7,7	55,3
	GBP 20,800 - GBP 25,999	216	7,2	7,2	62,6
	GBP 26,000 - GBP 31,199	195	6,5	6,5	69,1
	GBP 31,200 - GBP 36,399	127	4,3	4,3	73,4
	GBP 36,400 - GBP 39,999	120	4,0	4,0	77,4
	GBP 40,000 - GBP 44,999	110	3,7	3,7	81,1
	GBP 45,000 - GBP 49,999	128	4,3	4,3	85,3
	GBP 50,000 - GBP 59,999	118	4,0	4,0	89,3
	GBP 60,000 - GBP 74,999	121	4,1	4,1	93,3
	GBP 75,000 - GBP 99,999	110	3,7	3,7	97,0
	GBP 100,000 or more	89	3,0	3,0	100,0
	Total	2987	100,0	100,0	

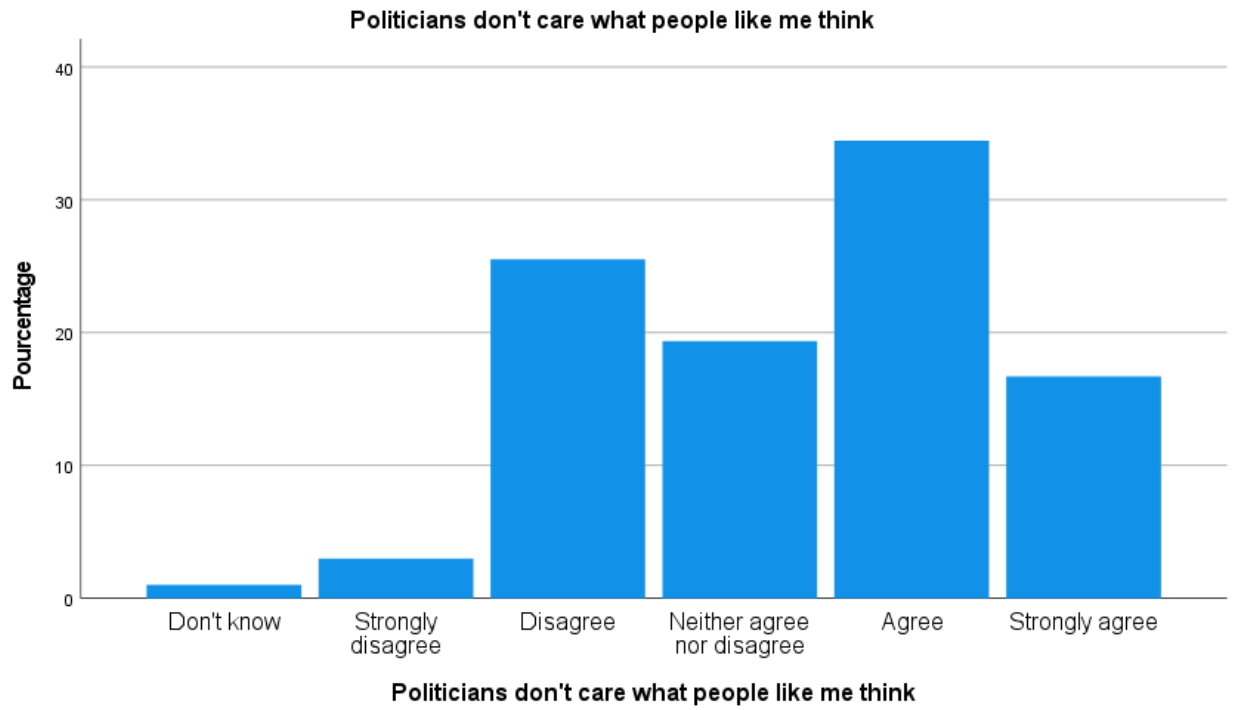


From the analysis of the graph and table, we can say that most respondents are not comfortable with disclosing their income level and 11.1% of respondents have an income between GBP 10,400 - GBP 15,599.

Only 3% have an income level above GBP 100,000 and 1.7% for people with an income level below GBP 2,600.

Politicians don't care what people like me think

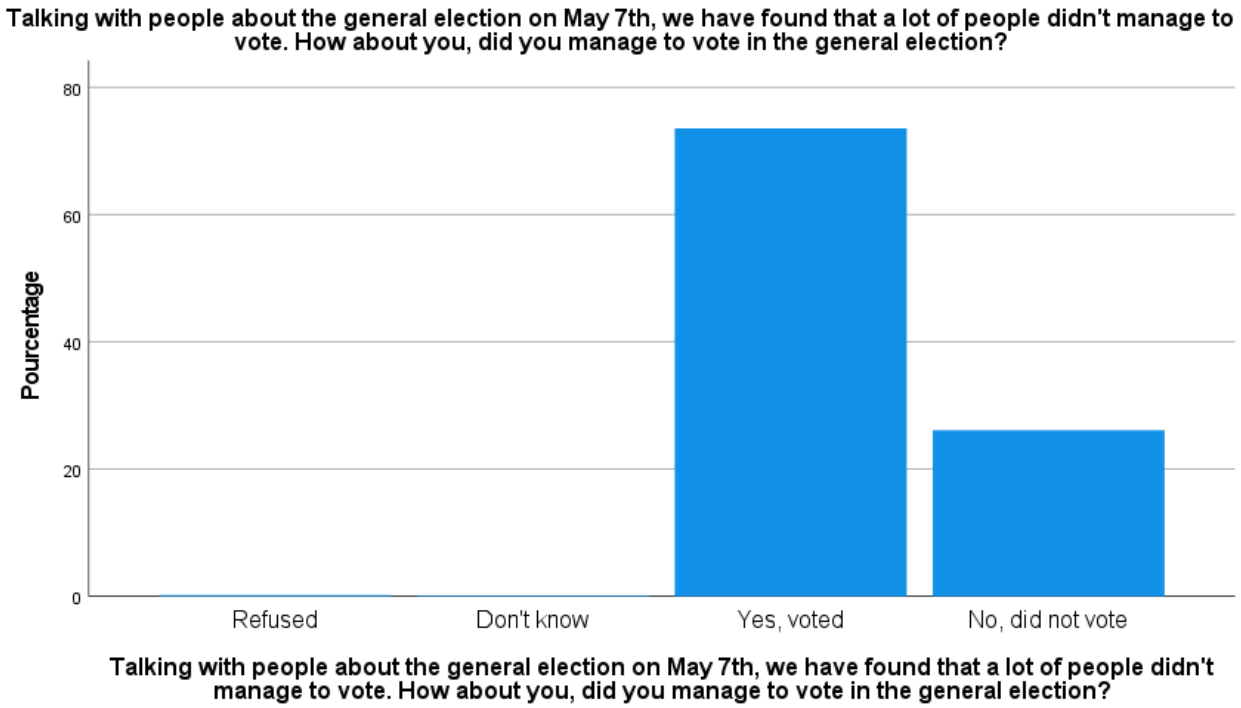
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Don't know	30	1,0	1,0	1,0
	Strongly disagree	89	3,0	3,0	4,0
	Disagree	762	25,5	25,5	29,5
	Neither agree nor disagree	578	19,4	19,4	48,8
	Agree	1029	34,4	34,4	83,3
	Strongly agree	499	16,7	16,7	100,0
	Total	2987	100,0	100,0	



34.4% of respondents agree that politicians do not care what they think, which indicates a fairly low level of satisfaction on the part of politicians, and 16.7% totally agree.

Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election?

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Refused	6	,2	,2	,2
	Don't know	4	,1	,1	,3
	Yes, voted	2197	73,6	73,6	73,9
	No, did not vote	780	26,1	26,1	100,0
	Total	2987	100,0	100,0	



Despite this high rate of dissatisfaction with political parties, 73.6% of respondents voted in the May 07 elections.

On the other hand, 26.1% of respondents did not vote, and abstention rates were very low.

Question II :

Hypotheses:

H_0: The variables "Politicians don't care what people like me think" and "Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election?" are independent.

H_a: The variables "Politicians don't care what people like me think" and "Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election?" are dependent.

a) Our dependent variable is the one with the wording "Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election?" and the independent variable is the one labelled "Politicians don't care what people like me think".

b) Let's create a cross tab

Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election? * Politicians don't care what people like me think Crosstabulation

Count

		Politicians don't care what people like me think						
		Don't know	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree	Total
Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election?	Refused	0	0	0	0	2	4	6
	Don't know	0	0	0	1	0	3	4
	Yes, voted	13	79	649	435	718	303	2197
	No, did not vote	17	10	113	142	309	189	780
Total		30	89	762	578	1029	499	2987

We note that 718 respondents think that politicians don't care about them, but still voted in the May 07 elections. And 113 think the opposite but were unable to vote.

c) Chi-square test for independence

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	144,993[a]	15	<,001
Likelihood Ratio	146,092	15	<,001
Linear-by-Linear Association	41,969	1	<,001
N of Valid Cases	2987		

a. 12 cells (50,0%) have expected count less than 5. The minimum expected count is ,04.

With an α of 0.05, Pearson Chi-square of 144.993 and p-value < 0.01, we reject the null and state that we have sufficient evidence to support that two variables are dependent.

d) The p-value<0.01, the strength of association between these two variables is highly significant at an α threshold of 0.05.

Question III :

Hypotheses:

H_0: Income levels and attitude to voting are independent

H_a: Income levels and attitudes to voting are dependent on

a) The dependent variable is voting attendance, the variable labelled "Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election?" and the independent variable is income level.

b) Let's create a cross tab

Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election? * Income - gross household (represents the total income of your household from all sources before tax - including benefits, savings and so on?) Crosstabulation

Count

		Talking with people about the general election on May 7th, we have found that a lot of people didn't manage to vote. How about you, did you manage to vote in the general election?				
		Refused	Don't know	Yes, voted	No, did not vote	Total
Income - gross household (represents the total income of your household from all sources before tax - including benefits, savings and so on?)	Refused	1	1	309	106	417
	Don't know	1	0	221	111	333
	Under GBP 2,600	0	0	25	27	52
	GBP 2,600 - GBP 5,199	1	0	51	37	89
	GBP 5,200 - GBP 10,399	1	0	124	75	200
	GBP 10,400 - GBP 15,599	0	0	239	92	331
	GBP 15,600 - GBP 20,799	1	0	161	69	231
	GBP 20,800 - GBP 25,999	0	1	161	54	216
	GBP 26,000 - GBP 31,199	0	0	147	48	195
	GBP 31,200 - GBP 36,399	0	2	103	22	127
	GBP 36,400 - GBP 39,999	0	0	87	33	120
	GBP 40,000 - GBP 44,999	0	0	90	20	110
	GBP 45,000 - GBP 49,999	0	0	103	25	128
	GBP 50,000 - GBP 59,999	0	0	100	18	118
	GBP 60,000 - GBP 74,999	0	0	99	22	121
	GBP 75,000 - GBP 99,999	0	0	97	13	110
	GBP 100,000 or more	1	0	80	8	89
Total		6	4	2197	780	2987

27 people with an income below GBP 2,600 did not vote in the May 07 elections and 80 of those with an income above GBP 100,000 did.

c) Chi square test

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	139,550[a]	48	<,001
Likelihood Ratio	129,686	48	<,001
Linear-by-Linear Association	40,302	1	<,001
N of Valid Cases	2987		

a. 34 cells (50,0%) have expected count less than 5. The minimum expected count is ,07.

With an α of 0.05, Pearson Chi-square of 139.550 and p-value < 0.01, we reject the null and state that we have sufficient evidence to support that two variables are dependent.

d) The p-value<0.01, the strength of association between these two variables is highly significant at an α threshold of 0.05.

e) Both tests have a p-value<0.01, But looking at the Pearson Chi-Square values, which are 144.993 for the test on satisfaction with politicians and 139.550 for the test on income level, we can conclude that respondents' level of satisfaction with politicians has a greater impact on voting behaviour. We can conclude that respondents' level of satisfaction with politicians has a greater impact on voting behavior.

Question IV:

a) Hypotheses:

H_0: All variables are independent of each other.
H_a: Not all variables are independent. There is at least one association that is dependent on.

b) Let's create a cross tab

- 12 respondents with an income level below GBP 2,600 and agreeing that politicians don't care what they think agreed to vote on May 07.

- 45 respondents with an income above GBP 100,000 and who don't think that politicians don't care about them, voted on May 07.

c) With Pearson Chi square of 144.993 and p-value < 0.01, we conclude that not all variants are independent and that there is at least one significantly dependent association.

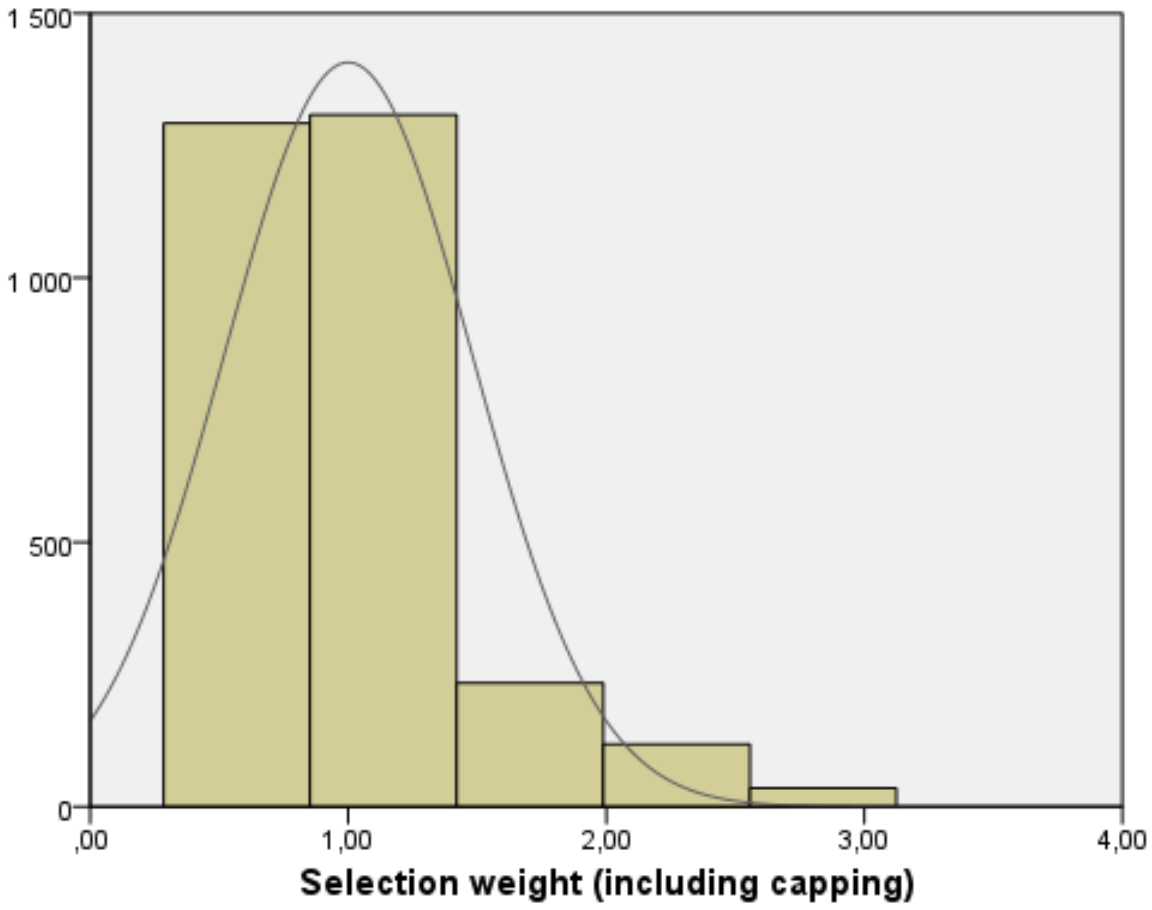
d) By considering each modality of the variable representing the level of income only at the "Don't Know" level, we are able to reject the H0 hypothesis of non-dependence, since p-value<0.01. We can therefore conclude that the variables are significantly dependent, since each of the modalities influences the dependence.

NB: Three-way crosstabs and chi-square tests are available in the .spv source file.

Question V:

The selected scale variable is "Selection weight (including capping)". This variable is chosen because it is a scale variable and therefore a quantitative variable.

a) Distribution



This is an asymmetrical distribution and most selection weights are between 0.5 and 1.4.

Selection weights between 2.0 and 3.0 represent a minority.

b) Descriptive Statistics

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation	Variance
Selection weight (including capping)	2987	,57	2,84	1,0000	,48112	,231
Valid N (listwise)	2987					

The mean value is 1.0, with a standard deviation of 0.48112 and a variance of 0.231.

Interpretation

- The average weight of the selection is 1.0
- The smallest selection weight is 0.57 and the largest is 2.84
- With a standard deviation of 0.48112, we can conclude that the values are clustered around the mean.